

Benchmark for Crack Re-Identification in Ticket Management Systems

Délcio Ferramenta, Cátia Loureiro, António Silva, Beatriz Miranda,
Paulo Brito, Nuno da Costa, Ricardo Rodrigues, Duarte Fernandes
DTx CoLAB / University of Minho, Campus de Azurém,
Guimarães, Portugal
delcio.ferramenta@dtxcolab.pt

Abstract—Building ticket management systems must decide whether a newly reported defect matches an existing ticket or needs a new ticket. Repeated reports of the same crack from different viewpoints fragment history and duplicate assessment work. We study this visual association as crack re-identification (re-ID) and introduce CRACKID (140 photographs, 17 walls, 212 identities in the frozen benchmark snapshot: 6 validation walls and 11 test walls with 19 matchable test identities). We evaluate ten zero-shot crop-based matchers from four families under retrieval, thresholded pairwise decisions, and open-set identification with rejection. Retrieval varies (Rank-1 0.370–0.613), but validation-calibrated pairwise decisions cluster just above the best-constant floor (pairwise F_1 at validation threshold $F_1^{\text{@v}}$ 0.331–0.338 versus 0.305 floor; oracle F_1^* 0.337–0.374). Context and crack-erasure controls implicate wall location over crack appearance. A geometric reference baseline that registers wall texture with cracks masked out and compares projected masks achieves a detection and identification rate (DIR) of 0.768 at 0.1 false alarm rate (FAR) on the 48% alignable subset, versus 0.384 for the best crop method, 0.404 for the best context control, and 0.000 for a coverage-only control. All figures are point estimates on a small wall-clustered sample with single-annotator homography-propagated labels, so the open-set advantage is provenance-conditioned and ranking measures short-baseline within-visit correspondence. This establishes a dataset and benchmark for the visual association step; cross-visit evaluation and ticket-level duplicate reduction remain future work.

Index Terms—Annotation Quality, Benchmark, Building Maintenance, Crack Re-Identification, Geometric Alignment, Homography, Structural Inspection

I. INTRODUCTION

Ticket management systems organize physical-defect reports into records for inspection, prioritization, intervention, and follow-up. Each ticket should hold the history of one incident, yet the same crack may be photographed repeatedly by different users, creating duplicate tickets that fragment history and duplicate assessment work. Each new observation must then be linked to its ticket, or a new ticket created for unrecorded damage.

This is visual re-identification (re-ID) of photographic evidence: (i) unchanged morphology under varying viewpoint, distance, field of view, illumination, or partial visibility; and (ii) changed damage (extension, widening, branching) whose identity must persist, distinguishing growth from new damage. Errors cause duplicates or misattribution.

Current crack methods focus on single-image detection and segmentation. Datasets such as CrackSeg9k [1] and Omni-Crack30k [2] lack cross-observation identities, so observations cannot be linked to tickets. Whether person/object re-ID [3], [4] transfers to cracks—where distinct defects look alike and one defect varies widely—is untested.

We introduce CRACKID to decide whether photos show the same wall crack: 140 photos of 17 walls from one visit (6 validation, 11 test; 19 matchable test identities; $n_Q=305$). Real photos cover single-visit acquisition change; appearance and growth are probed only synthetically. Cross-visit performance with real photos is future work. All accuracies are point estimates on a small wall-clustered sample.

The benchmark evaluates ten zero-shot matchers on CRACKID, separating candidate retrieval from thresholded same-defect decisions and open-set identification with rejection: high rank alone must not attach a report. Context and crack-erased controls separate crack from wall-location cues, and a geometric baseline aligns walls and compares projected masks.

We address Research Question (RQ)1 (association under acquisition change), RQ2 (context and geometric alignment), and RQ3 (synthetic appearance/growth change); RQ1–RQ2 use real single-visit photos, RQ3 is exploratory synthetic. Ticket-level duplicate reduction and cross-visit evaluation are future work.

Contributions: (i) cross-observation identities for CRACKID with stated annotation provenance (Sec. III-B); (ii) a benchmark splitting retrieval, same-defect decisions, and open-set identification (Secs. IV and VI); (iii) context controls, geometric reference, and coverage analysis (Secs. V, VI-C, and VII-B); (iv) a synthetic revisit probe (Sec. VII-C).

II. RELATED WORK

Crack analysis targets single-image detection and segmentation [5]–[7]; consolidation studies keep that scope [1], [2], and multi-view pipelines fuse one inspection into a surface or segmentation [8] without instance correspondence. Structural-health monitoring tracks cracks over time with fixed cameras and marked locations [9]–[12], where identity is *given by construction*: unlike our open-set association task, they measure a known crack.

Our setting is object re-ID [4] with open-set identification [13]: Scale-Invariant Feature Transform (SIFT) [14] and Oriented FAST and Rotated BRIEF (ORB) [15], learned correspondences including SuperGlue [16] and LoFTR [17], and region embeddings [3], [18]–[21] are all evaluated here zero-shot without CRACKID fine-tuning (YOLOv8 row: layer-9 `yolo_v8n.pt` features [22]). LightGlue [23] and NetVLAD [24] are noted as future retrieval baselines. Vehicle-damage re-ID across claims [25] shares our open-set protocol; related work targets detection and severity, not association [26]–[28]. Align-then-reason about presence/absence is change detection [29], [30] with planar registration [31]–[33] and Enhanced Correlation Coefficient (ECC) fallback [34]; our point is that alignment and matching targets must be *disjoint*, with the homography from the wall while the crack is identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID frozen benchmark snapshot has 140 photographs of 17 exterior walls taken at one site over four half-days with two consumer handheld devices (Mendeley Data [35]; Fig. 1). Walls are stippled render (textured) or smooth painted plaster where cracks are the only visible features. The split is six validation walls (walls 01–06; 66 photos, 66 identities) and eleven test walls (walls 07–17; 74 photos, 146 identities). Only identities in at least two photographs are answerable queries (18 validation, 19 test; $n_Q=439$ validation and 305 test). Walls 18–19 were added after freezing and are excluded from all results. Each wall is a continuous walk-around (median 3 s between frames, max 81 s); data are within-visit viewpoint and scale correspondences, an upper bound on multi-visit performance.

Adjacent-frame retrieval. All 305 answerable test queries include the adjacent frame in gallery. Gap sweeps in Sec. VII-B (305 to 205 to 149 to 105 queries for gaps 0–3) retain Rank-1 as the pool shrinks, indicating robustness to short baselines.

B. Annotation, and why it needs auditing

Oriented FAST and Rotated BRIEF (ORB) homography propagates single clicks frame to frame; fragments are then merged into physical-crack identities. The merge is the main risk (a wrong merge turns a hard negative into a false positive). In the frozen snapshot, 95 of 212 identities (45%; 70 of 146, 48% in test) come from merges, with median bridged gaps of 228 pixels (test) and 286 pixels (validation). One annotator checked all clicks and triaged unclaimed points (107 of 1,023 as no crack; details archived with the code). Single-annotator triage reduces but does not remove merge risk. All counts refer to the frozen snapshot (Table I); the live `dataset/labels/` directory adds walls 18–19 and must not be mixed with frozen figures without rerunning the benchmark.

IV. BENCHMARK PROTOCOL

The benchmark measures visual association: given a new crack observation, does it match a recorded defect? Annotated

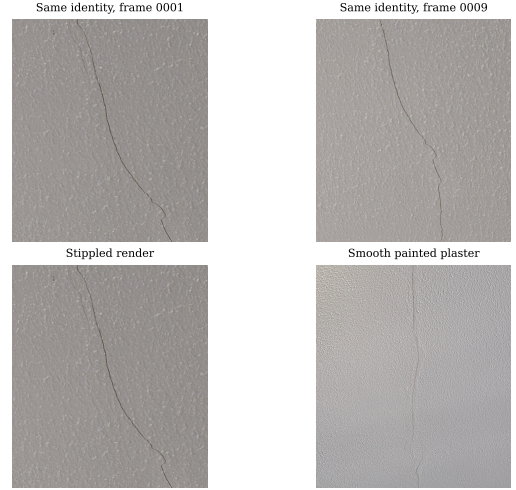


Fig. 1. Real CRACKID examples. Top: one identity across non-adjacent frames. Bottom: stippled render versus smooth plaster. Labels via homography propagation, unaudited.

TABLE I
COUNTING UNITS PER METRIC (FROZEN SNAPSHOT, $n_Q=305$ TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	305
Positive query–gallery pairs	26 012	5 722
Valid query–gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

identities stand in for permanent defect records, so scores measure association accuracy, not ticket-level duplicate reduction.

A. Task formulation and candidate records

A wall is one physical wall; an identity links observations of the same crack (Sec. III-B). A query is one crack instance; its gallery holds other instances from photos of the same known wall. Cross-wall search is excluded. A query is *known* if its gallery contains the identity, else *unknown*. Each method scores query–gallery pairs for ranking, thresholded decisions, and identification with rejection. Pairs are the unit for pairs; image pairs for registration/assignment. The Hungarian one-to-one constraint applies per image pair only.

Counts for the frozen snapshot (6 validation walls 01–06; 11 test walls 07–17; walls 18–19 excluded) are in Table I. Registration covers all 301 same-wall test image pairs; 255 yield scored pairs with validity-mask cells (Sec. VII-B).

B. Test conditions, decisions, and metrics

The main test uses real walk-arounds per wall with morphology assumed unchanged; variation is viewpoint, distance, scale, lighting, and visibility, plus same-wall distractors and unknown queries. Validation/test walls are disjoint. All methods run as-is on shared masks without CRACKID tuning (Sec. V); thresholds fit on validation apply to test: within-visit

correspondence only. A synthetic check separates photometric (condition B: blur/noise/contrast with unchanged structure) from growth effects (condition A: one tip extended) on ten photos from five walls with one seed and a common warp (parameters archived with code), comparing OSNet with context against HYBRID-SKEL ($\tau=8.0$ px), distinct from REG+CHAMFER: a preliminary check only.

Decisions: pairs above threshold are accepted; per query only the top gallery entry above threshold is accepted; per image block Hungarian assignment applies with below-threshold drops. Unscorable pairs are kept (predicted negative for F_1 , last rank for retrieval); registration failure is lack of evidence, not evidence of a new defect. Metrics are Rank-1 and mean Average Precision (mAP); pairwise F_1 with oracle F_1^* versus validation F_1^{val} ; assignment $F_1^a=2\text{TP}/(2\text{TP}+\text{FP}+\text{FN})$; open-set detection and identification rate (DIR) and false alarm rate (FAR) with DIR at FAR= 0.1; plus coverage (fraction scored). References are random ranking and best-constant F_1 $2p/(1+p)$ (0.305 at prevalence 0.180; 0.544 on the 0.48 geometric subset; 0.452 validation at 0.292); full chance levels in Table II. With 11 walls and 19 matchable identities we report point estimates (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack locations on the wall drive the association: it aligns wall surfaces to a common frame and then compares the projected crack masks. It assumes a roughly planar wall and uses no CRACKID training. The shared masks (trained on public crack data only; no CRACKID wall in training) use instance extraction at 200 px and 5 px to keep inputs consistent across methods. Masks are taken as given, and their error is not measured.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from the surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and the succeeding stage, not guaranteed per pair. Full detector settings (Scale-Invariant Feature Transform (SIFT) with Contrast Limited Adaptive Histogram Equalization (CLAHE) [36], ratio test, robust estimation) are archived with the code.

B. Registration cascade and validity checks

Textured render generally provides background features, whereas smooth plaster does not. A three-stage cascade is used: masked SIFT; if too few keypoints remain, unmasked SIFT as fallback (flagged per pair, since the crack then aids alignment); finally Enhanced Correlation Coefficient (ECC) photometric alignment. Candidates pass plausibility tests (no mirrored or folded quads; scale within $[1/16, 16]$); the surviving stage is recorded per pair, and its inlier root-mean-square error (RMSE) sets the comparison tolerance. Unscored pairs count as lack of evidence, not evidence of different defects.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance in (1), which tolerates the small misalignments that collapse intersection over union (IoU) on narrow cracks. Let a, b be crack point sets and $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right]. \quad (1)$$

Early rejection of distant candidates applies. Chamfer coverage (fraction within tolerance τ of the per-pair RMSE, clamped to $[2, 16]$ pixels, default 6 pixels; HYBRID-SKEL uses 8.0 pixels) is evaluated separately as its scale-free counterpart (see Sec. VII-B).

D. Correspondence decisions and limits of spatial evidence

For each image pair, one-to-one assignment follows the Hungarian method [37] with the validation threshold shared by all methods; unscored pairs take value $-\infty$ (predicted negative, ranked last, as in Sec. IV). An aligned projection onto visibly empty wall is a *licensed negative*—evidence of absence, stronger than low embedding similarity—while missed detections, out-of-frame areas, and registration failures stay unresolved. The REG+CHAMFER reference targets unchanged damage; progression uses HYBRID-SKEL. Performance depends on surface-dependent failures and on homography-based annotation provenance.

VI. THE APPEARANCE CEILING

A. Setup

Ten methods use crack-centered crops and a common validity mask: SIFT and ORB with Random Sample Consensus (RANSAC) inliers, SuperGlue and LoFTR with pretrained indoor weights, five embedding methods, and re-ID OSNet with cosine similarity, all zero-shot on CRACKID; YOLOv8 uses layer-9 `yolo8n.pt` features. Input sizes and checkpoints are archived with the code. Context controls with fixed backbones (OSNet, CLIP): `@ctx1` enlarges the instance box by $1 \times$ per side; `@ctx1@nocrack` inpaints the crack to isolate wall context. REG+CHAMFER aligns walls then compares projected masks (Sec. V); COVERAGE-ONLY scores registered pairs constantly without looking at cracks. Synthetic tests compare OSNet with context against HYBRID-SKEL variants.

B. Result: appearance buys almost nothing on the decision

Table II (11 walls, frozen snapshot, $n_Q=305$) shows Rank-1 0.370–0.613 (best ViT-B/16; best mAP 0.497 OSNet; best crop DIR at FAR= 0.1 0.384 SuperGlue), yet oracle F_1^* 0.337–0.374 and validation-calibrated F_1^{val} 0.331–0.338: ranking does not imply deciding. The narrow F_1^{val} band shows poor threshold transfer across walls. With 19 matchable test identities we claim no fine-grained crop ranking. All values are point estimates.

TABLE II

MAIN RESULT ON CRACKID TEST (11 WALLS, FROZEN SNAPSHOT, $n_Q=305$). F_1^* USES AN ORACLE TEST THRESHOLD; ΔF_1^* IS EXCESS OVER THE FULL-POOL FLOOR; $F_1^{\text{@}v}$ USES THE VALIDATION THRESHOLD. BOLD IS BEST CROP PER COLUMN. GEOMETRY COVERS 0.48 WITH FLOOR 0.544, SO ITS ΔF_1^* IS NOT COMPARABLE TO CROP ROWS (SEC. VII-B). S/QUERY IS TOTAL-S/305 ON ONE WORKSTATION (RELATIVE TIMES ONLY).

Method	Family	R@1	mAP	DIR @FAR= 0.1	F_1^*	ΔF_1^*	$F_1^{\text{@}v}$	cov.	s/q
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [22]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

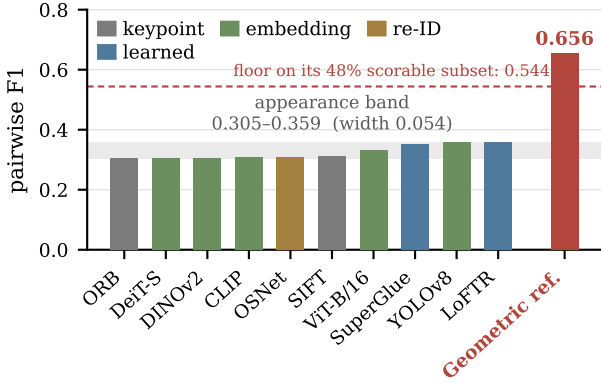


Fig. 2. Appearance ceiling: pairwise F_1^* stays narrow while ranking spreads. Geometry (red) leaves the band but uses the 0.48 pool with floor 0.544 (Table II).

C. Is it the crack, or the wall behind it?

Context helps ranking and open-set performance but most gain remains without the crack: OSNet Rank-1 0.603 to 0.775 (0.711 crack-erased), DIR at FAR= 0.1 0.154 to 0.404 (0.393 erased); CLIP 0.143 to 0.311 (0.286 erased). Both erased views beat all tight crops, implicating wall location as the cue. Pairwise F_1^* stays 0.307–0.361 across base, context, and erased views, while open-set gain is +0.250 for OSNet. Within-visit retrieval works, but reliable automatic decisions do not; ranked lists may assist operators, while acceptance needs more evidence (Fig. 2).

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometry achieves Rank-1 0.952 versus 0.613 best crop (ViT-B/16) and mAP 0.886 versus 0.497 (OSNet), at 0.48 versus 1.00 coverage. Above COVERAGE-ONLY, ranking gains persist but the open-set gain does not: DIR at FAR= 0.1 is

0.768 versus 0.384 (SuperGlue), 0.404 (best OSNet+ctx1), and 0.000 for COVERAGE-ONLY (chance 0.038). For hairlines crop similarities rarely reject, while aligned empty projections are valid negatives. Cost is 7.19 s/query (LoFTR) versus 0.006 s/query (OSNet); geometry needs 27.9 s/query and 26.0 s/pair (relative only; SuperGlue/LoFTR timed separately). For RQ2, alignment enables identification beyond coverage, conditioned on success, surface, and provenance. Point estimates on 19 identities.

B. Coverage, failures, and ablations

Crops score 100% of pairs; geometry 48%. Unsourced pairs take negative score and last rank. Censoring penalizes ranking but inflates pairwise F_1 : scored sets keep 100% of positives but 36.8% of negatives (prevalence 0.180 to 0.373; floor 0.305 to 0.544). Coverage-only is 0.544 F_1 , 0.533 mAP, 0.515 Rank-1, so much ranking/pairwise strength is *which pairs align*; only the open-set lead (0.768 vs. 0.000) is not. Failures are wall-dependent (zero on two test walls; per-wall table with code). Adjacent frames answer all 305 queries, so ranking is short-baseline; gap sweeps (305 to 105 at gap 3) hold Rank-1. Registration follows sharpness, not frame gap (Wald $z=5.11$, $p=3.3 \times 10^{-7}$ vs. -2.78 , $p=5.4 \times 10^{-3}$); sharpness ≥ 10 gives 56.1% registration on the filtered set minus one wall. Scale-free coverage costs 0.240 mAP and 0.103 Rank-1 versus Chamfer (DIR +0.014). Disabling crack exclusion leaves ranking unchanged but raises DIR +0.057; masking is kept for independence. ECC adds ~ 0.03 Rank-1, 0.02 mAP. Total 26.0 s/pair (7817.3 s over 301 pairs; 27.9 s/query in Table II), $\sim 3 \times$ LoFTR and $>100 \times$ the cheapest embedding.

C. A synthetic revisit protocol

Ten source photos from five walls (one seed) undergo the same small warp under (A) single-tip growth or (B) photometric corruption, then match real images (procedure with code;

TABLE III
HYBRID-SKEL VERSUS OSNET+CTX ON SYNTHETIC REVISIT; F_1 AT
VALIDATION THRESHOLD. ONE SEED, FIVE WALLS. HYBRID-SKEL
DIFFERS FROM REG+CHAMFER IN TABLE II.

Protocol	Method	n_q	R@1	mAP	DIR @FAR _{0.1}	$F_1^{\text{@v}}$	F_1^a
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.631
B (appear.)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609

p90 tilt 75°). HYBRID-SKEL differs from REG+CHAMFER in Table II. Table III compares HYBRID-SKEL and OSNet+ctx.

For ranking/open-set OSNet+ctx wins (0.826/0.671 and 0.739/0.663 vs. 0.275/0.274 and 0.343/0.342), while HYBRID-SKEL leads F_1^a (0.646/0.640 vs. 0.631/0.609) with similar $F_1^{\text{@v}}$ (0.569/0.574 vs. 0.601/0.594); no cell from pure homography ($H=0.0\%$). Ordering reverses by metric (RQ3): Table II does not imply multi-visit robustness.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This study gives the first measurable single-visit crack re-ID test with a geometric reference. Results rest on a small wall-clustered sample with single-annotator labels and partial registration coverage; each item below pairs the finding with the step to verify it.

- 1) **Scale to more walls and visits.** Collect more walls, surface families, and annotators for wall-level confidence intervals and reliable ranking. Above all, add a second session at deliberately different distances and lighting to turn within-visit correspondence into a true cross-visit ticket-backend benchmark.
- 2) **Add provenance-free annotation.** One annotator checked all clicks and triaged unclaimed points, but agreement is unmeasured and the geometric lead is provenance-conditioned. A small homography-free label subset plus released per-pair score matrices for sensitivity, wall-clustered intervals, and threshold-free average precision (AP)/area under ROC (AUROC) would isolate the provenance-independent open-set gap.
- 3) **Close the coverage gap.** The baseline answers 48% of pairs (97 of 301 image pairs; 15,328 of 31,822 valid cells) and scores zero on smooth plaster. Surface-adaptive registration—learned local features, direct photometric alignment, or sparse depth from consumer devices—would approach full evaluation and further separate geometry from homography-based provenance.
- 4) **Learn from geometric supervision.** The baseline is an unlearned reference, not a deployed system. Pseudo-labels from registration or fine-tuning a foundation backbone on homography correspondences would keep the location advantage without per-pair registration.
- 5) **Assess the ticket lifecycle.** Define task metrics—duplicate rate and missed-new-defect rate—and test seg-

mentation, re-ID, and joint ticket assignment in closed-loop simulation of real inspections.

Together these steps turn the within-visit reference into a deployable multi-visit re-ID component for operational tickets.

IX. CONCLUSION

We present CRACKID for testing crack re-ID: 140 photos from 17 walls with 212 identities in the frozen snapshot (6 validation, 11 test; walls 18–19 excluded); annotation provenance is stated (45% merged, 48% of test from merges) and unclaimed-click triage is complete (107 of 1,023 as no crack). All accuracies are point estimates without wall-level intervals (19 matchable test identities, $n_Q=305$); we assert the pattern, not the digits.

The headline result is negative and structural: when decisions are required, ten zero-shot matchers from four families sit +0.032 to +0.069 above the 0.305 best-constant floor (F_1^* 0.337–0.374), with validation-calibrated decisions at $F_1^{\text{@v}}$ 0.331–0.338. Retrieving from a gallery is distinct from deciding. The limit is not crack appearance alone: a geometric reference that registers wall texture with cracks masked and Chamfer-matches projected masks achieves DIR at FAR= 0.1 of 0.768 versus 0.384 best crop and 0.404 best context on 48% of pairs, while a COVERAGE-ONLY control using the same alignment without the crack scores 0.000 open-set (pairwise F_1 0.544 on its own 0.544 floor).

Operationally, the open-set gain is a *licensed negative*—positive evidence of absence—where crop embeddings show only low similarity; whether it is independent of homography-based annotation is unproven and it must be treated as provenance-conditioned. Two gaps remain: a real second session to make the benchmark cross-visit, and a second annotator with a homography-free audit subset for measured agreement. Coverage of 48% is itself a deployment shortfall: smooth plaster can be unanswerable. Finally, the small single-seed synthetic revisit reverses the headline ranking in favor of OSNet+context, with distinct HYBRID-SKEL keeping only a small assignment- F_1 lead; this is a within-visit geometric reference, not a multi-visit ticket-backend solution.

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AI disclosure: Generative AI was used only for minor language editing. All methods, results, figures, and analyses are original and the authors take full responsibility for their accuracy.

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